Graphic Era HILL UNIVERSITY
Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)
University under section 2(f) of UGC Act, 1956

End Term (Odd) Semester Examination December 2024

Roll no.....

Name of the Course and semester: B Tech CSE Sem 7th
Name of the Paper: Artificial Intelligence
Paper Code: TCS 706
Time: 3 hour

Maximum Marks: 100

Note:

(i) All the questions are compulsory.
(ii) Answer any two sub questions from a, b and c in each main question.
(iii) Total marks for each question is 20 (twenty).
(iv) Each sub-question carries 10 marks.

Q1 (10 marks) CO1

Define Artificial Intelligence (AI). Explain in detail how AI is applied to problem-solving in games, natural language processing, automated reasoning, and visual perception. Provide examples to illustrate your answer.

(b)
You are tasked with developing a heuristic algorithm for a simple game scenario where a player must reach the target score of 50 by selecting numbers from the set {1, 3, 5, 7}. The algorithm prioritizes larger numbers first. Simulate the algorithm's decision-making process to solve for the target score of 50.

(c)
AI often aims to simulate human-like intelligent behavior. Critically evaluate the challenges and limitations of replicating sophisticated tasks such as natural language understanding and visual perception. Provide examples to support your analysis.

Q2 (10 marks) CO2

(a)
Explain context-free grammar and transformational grammar. Construct a detailed parse tree for the sentence: "The cat sat on the mat." Specify the grammar rules used.

(b)
Given the transition net below, simulate the parsing process for the input sentence: "John runs quickly."
Transition Net:
S → NP VP
NP → {John, Mary} VP → V Adv
V → {runs, walks}
Adv → {quickly, slowly}
Show each transition step and the resulting structure.

(c)
Describe the concept of transformational grammar and its role in natural language processing. How does it differ from context-free grammar in handling complex sentence structures? Provide examples.

Q3 (10 marks) CO3

(a)Discuss the role of first-order predicate calculus in knowledge representation. Write expressions to represent the following facts and query their relationship:
- "All birds can fly."
- "Penguins are birds but cannot fly."
- "Tweety is a penguin." Query: Can Tweety fly?

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(b)
Write a PROLOG program to represent the family tree below Relationships:
- John is Mary's father.
- Mary is Tim's mother.
- Query: Who is Tim's grandfather?

Provide the output when the query is executed.

(c)
Compare and contrast semantic networks and production rules as methods for knowledge representation. Discuss the strengths and limitations of each approach with relevant examples..

(10 marks) CO4

Q4
(a)
Explain the architecture of an expert system with an example. Discuss the role of components like the knowledge base, inference engine, and user interface in systems such as MYCIN.

(b)
Consider an expert system for vehicle diagnostics with the following rules
Rule 1: IF the engine does not start AND the battery is dead, THEN the problem is the battery.
Rule 2: IF the engine does not start AND the battery is fine, THEN the problem is the starter.
Rule 3: IF the engine starts AND there is smoke, THEN the problem is overheating.
Simulate the system to diagnose the issue when:
• The engine does not start, and the battery is fine.
• The engine starts, and there is smoke.

(c)
Expert systems like MYCIN and DENDRAL have been pioneers in their respective domains. Compare their architectures, functionalities, and the domains they serve. What were their limitations, and how can modern systems overcome these issues?

(10 marks) CO5

Q5
(a)
Define pattern recognition and explain its application in speech recognition. Using an example, describe the steps involved in recognizing a word from an audio input.

(b)
Using a binary classifier, classify the following data points relative to the decision boundary x=5x = 5x=5: Points: (2, Positive), (6, Negative), (8, Negative), (4, Positive), (5, Positive). Query: Classify point (7). Show the calculations and decision process for the classification.

(c)
Explain the role of symbolic and structured descriptions in pattern recognition. How do they facilitate object identification in AI systems? Provide examples to illustrate their application.